EXERCISE NO: 2.1

CREATE A 3D SOLID MODEL FROM ISOMETRIC DRAWING

AIM:

To create the given 3D model with the required dimensions using Autodesk Fusion 360.

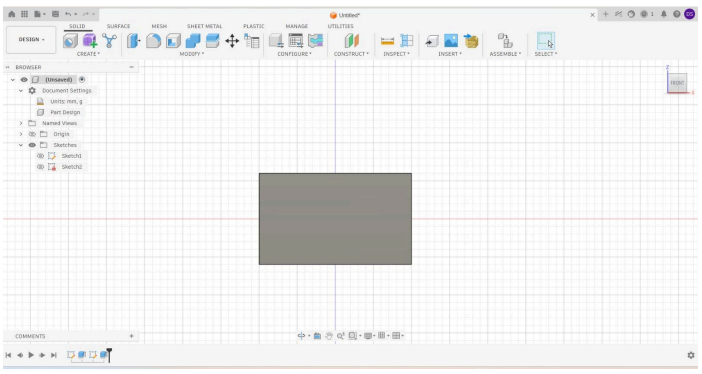
REQUIREMENTS:

- Autodesk fusion 360
- Windows 10
- 16 GB of RAM (integrated graphics recommend 1 GB or more)
- 2.5 Mbps or faster download; 500 kbps or faster upload

PROCEDURE:

Step 1: Select the Front View, click Create Sketch, and draw a Center Rectangle with a width of **60mm** and a height of **100mm**.

Step 2: Click Finish Sketch, select the rectangle profile, and Extrude it to a thickness of **30mm**.

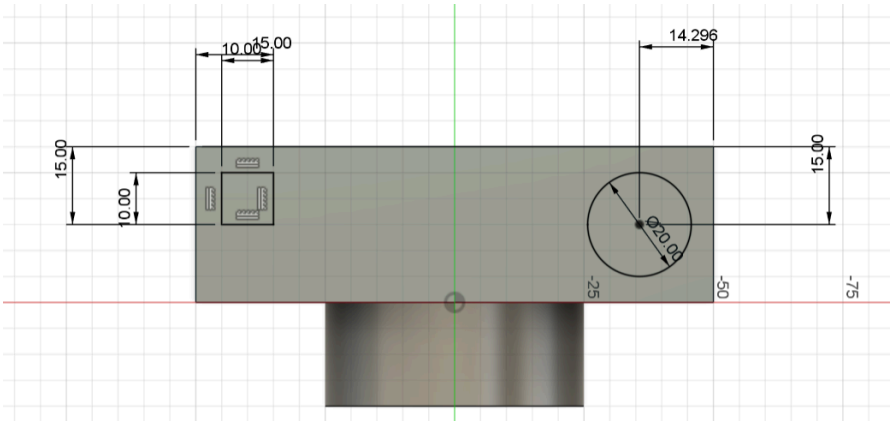


Step 3: Select the front face of the body, click Create Sketch, and draw a circle with a diameter of **50mm**.

Step 4: Click Finish Sketch, select the circle, and Extrude it with a thickness of **-20mm** using the Join operation.

Step 5: Select the Top View, click Create Sketch, and draw a circle of **20mm.**

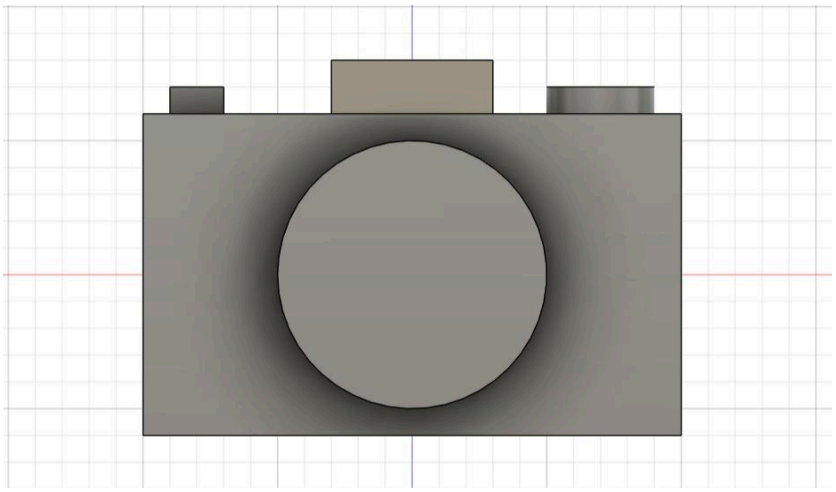
Step 6: Draw a rectangle, press the D key, and set the dimensions to **15*15mm**.



Step 7: Click Finish Sketch, select the shapes, and Extrude them upward by **5mm** using the **Join** option.

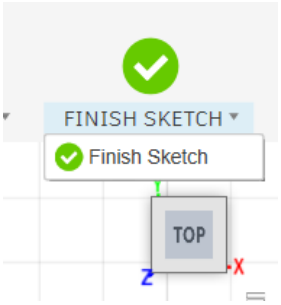
Step 8: Select the Back View, click Create Sketch, and draw a rectangle with dimensions of **30*35mm**.

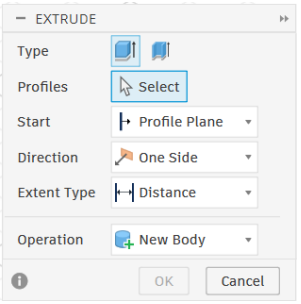
Step 9: Click Finish Sketch, select the rectangle, and Extrude it to a thickness of **-15mm**.



Step 10: Create a sketch on the side face, use the Line tool to draw the slope part at **45 degrees**.

Step 11: Click Finish Sketch and Extrude the slope profile with a thickness of **-30mm**.

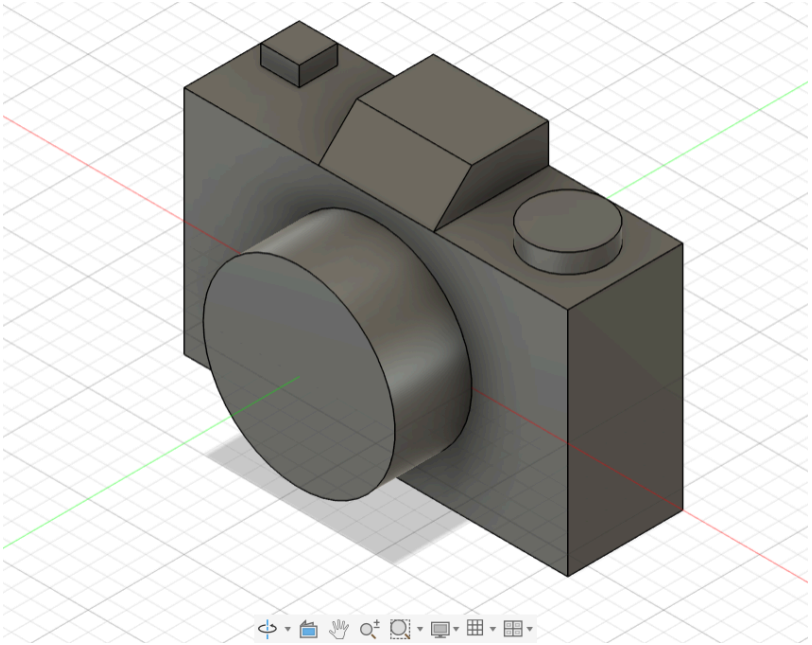




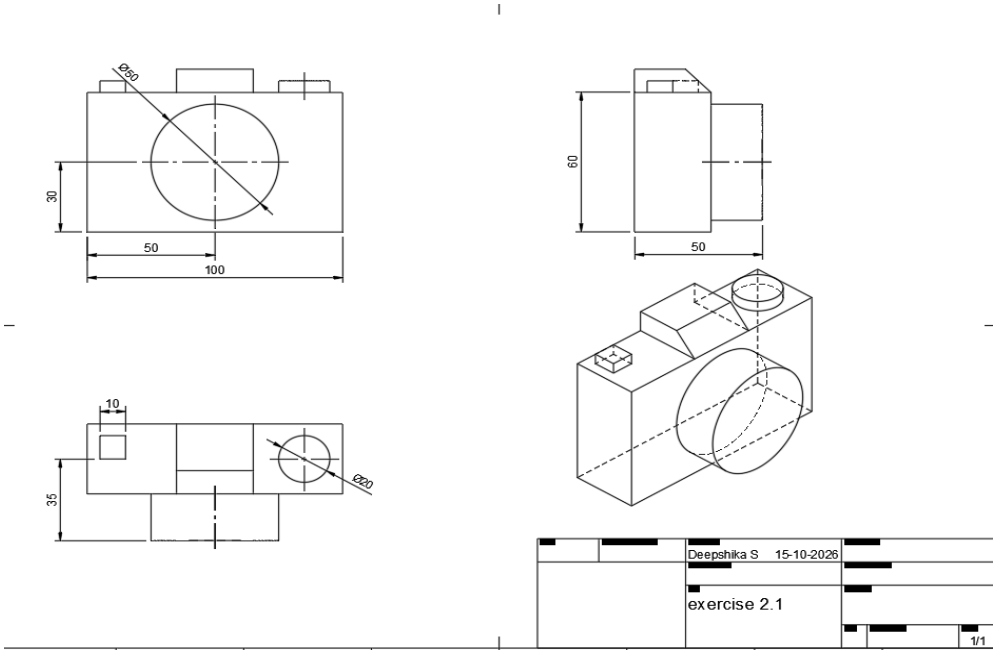
Step 12: Click the Save icon to save the file.

Step 13: Go to the menu options and select the Drawing option.

SKETCH:



OUTPUT:



RESULT:

Thus, the given 2D sketch was drawn and extruded using Autodesk Fusion 360.